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Research on the Method of Universal Forklift Pallet Detection and Pose Estimation Based on Visual and 3D Point Cloud Fusion

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***Abstract*—As logistics demands surge and labor costs rise, the industry turns to intelligent automation. Autonomous forklifts, key in this shift, autonomously transport goods to cut costs and boost efficiency. Pallet detection is undoubtedly a key step in achieving automatic forklifts. Traditional pallet detection methods, relying on single data modalities like images or 3D point clouds, face challenges with accuracy due to lighting, occlusion, and real-time processing. This paper introduces a novel approach for pallet detection and pose estimation on forklifts, merging visual data with 3D point clouds. Using a depth camera, the method refines pallet location via improved YOLOv8 detection and fuses color and depth data for precise 3D localization. The results show high performance and reliability, promising for real-world industrial applications.**

Keywords- Autonomous Forklifts; Visual Data; 3D Point Cloud; Pallet Detection; Pose Estimation

I. Introduction

In recent years, the logistics industry has experienced rapid growth, with forklifts emerging as indispensable material handling tools in warehousing, production lines, and logistics. The application of autonomous forklifts has increasingly gained attention in logistics automation. For these autonomous vehicles, the ability to quickly and accurately identify, locate, and estimate the pose of pallets in the working environment is crucial for enhancing the efficiency of logistics automation.

Pallets, as shown in Figure 1, come in a wide variety of shapes and sizes used for stacking and transporting goods. They can be categorized by the number of legs (two, three, four, or more), material (wooden, steel, plastic, etc.), color (blue, black, etc.), and fork entry position (center or off-center).

(a) (b) (c)

Figure 1. Different pallet types

With the rapid advancement of computer vision, 3D point clouds, and deep learning technologies, precise detection and pose estimation of forklift pallets have become feasible, offering new possibilities for improving the automation level and efficiency of forklift operations.

This study introduces a method that integrates visual data with 3D point clouds for the detection and pose estimation of pallets in autonomous forklift operations. It addresses the limitations of previous methods, which may not handle complex pallet variations and real-time operation demands effectively. Firstly, a depth camera is utilized to acquire both color and depth images of the pallet. Subsequently, the YOLOv8[1] detection algorithm from deep learning is applied to analyze the color images and extract features. Once the type, location, and center points of the pallet feet are identified from the images, this data is fused with the depth images to generate a 3D point cloud of the pallet, enabling precise localization. Then, the point cloud information is used to pinpoint the pallet feet, and pose estimation is performed based on the location of the feet. This process ultimately facilitates the automatic pallet handling task of the forklift. By combining visual and depth data, it precisely determines the pallet's position and orientation, which is crucial for forklift automation. This method is designed to be adaptable to various pallet designs and operational environments, offering a practical and cost-effective solution for enhancing the efficiency of forklift operations. The main contributions of this paper are as follows: first, it proposes a universal method for forklift pallet detection and pose estimation based on vision and 3D point cloud fusion, which achieves high accuracy; second, it validates the effectiveness and practicality of this method through experiments, demonstrating its potential application in forklift operations.

The paper is organized as follows: Section 2 reviews related work; Section 3 details the proposed method; Section 4 presents experiments and results; and Section 5 provides a summary and future outlook.

II. RELATED WORK

Pallet detection and pose estimation are essential for ensuring accurate loading and transportation of pallets by forklifts. This process involves using sensors to gather data and algorithms to determine the pallet's position and orientation within storage environments, where pallet arrangements vary, requiring adaptable and precise detection methods.

Sensor-based methods rely on sensor feedback for pallet localization. For example, Lecking et al.[2] used reflective markers on pallets detected by laser radar to outline the pallet and estimate its pose. However, these methods have limitations, such as wear and tear on the markers and narrow detection ranges with laser scanning, as noted by Bellomo et al.[3]. Baglivo et al.[4] combined point clouds with laser radar, but the time-consuming processing hinders real-time application. Mohamed et al.[5] applied machine learning to 2D laser radar data, achieving high detection rates but with less accurate pose estimation due to sparse laser data.

Traditional image-based approaches use image processing algorithms to exploit the shape and color attributes of pallets. Cui et al.[6] detected pallets based on color and edge information, which is limited to specific pallet types and struggles with multiple pallets. Xiao et al.[7] combined plane segmentation with template matching, but the method is sensitive to the quality of point cloud data. Kesuma et al.[8] used ArUco markers for real-time localization, but this approach incurs additional costs for marker installation and maintenance.

Deep learning-based methods have emerged as a cost-effective and robust alternative, training on pallet datasets to identify and localize pallets in images. Bohacs et al.[9] utilized YOLO9000[10] to detect specific pallet features and estimate pose, though limited to two-hole pallets. Mok et al.[11] employed multi-task classification for pallet positioning, but this method is tailored to specific scenarios. Zhu Danping et al.[12] improved upon CenterNet for pallet detection and introduced pose estimation, yet the method lacks detailed pose information due to limited keypoint data.

Despite the advancements, deep learning methods face challenges with complex scenarios like occlusions and lighting variations, and their computational demands limit widespread application. Enhancing the performance of deep learning in pallet detection and pose estimation remains a key area for research progress.

III. METHOD

In automated warehousing systems, forklifts are responsible for precisely picking up and transporting pallets. To achieve this, autonomous forklifts must first identify the pallet and then determine its tilt angle and distance to timely adjust the vehicle's position. Next, the forklift needs to align with the pallet's insertion holes, ensuring the pallet's center point is between the two forks, and adjust the vehicle to make the forks perpendicular to the pallet's surface, as shown in Figure 2. During the picking process, the forklift continuously locates the pallet to ensure the safe and smooth completion of the loading task. In summary, the three core functions required for an unmanned forklift to safely load pallets are: (1) detecting the pallet; (2) positioning the center point of the pallet; (3) calculating the pallet's tilt angle and distance.

Figure 2. The forklift aligns the pallet

Based on the analysis of common characteristics of different pallets, it is evident that relying on the end surface for pose estimation presents challenges when dealing with irregular pallets, as it may not provide accurate pose information. Additionally, the end surface does not offer information on the fork entry position, which is crucial when handling pallets with off-center forks. The position of the pallet feet becomes particularly important in such cases. Through comprehensive analysis, we have determined that the pose of any type of pallet can be translated into determining the pose of its feet. Therefore, we have designed a solution based on the YOLOv8-pose deep learning model, which first detects and recognizes the model in RGB images, then extracts point cloud ROI (Region of Interest), and finally estimates the pose by identifying the pallet feet. The overall scheme is divided into two stages: pallet recognition and pose estimation, as illustrated in Figure 4.

A. YOLOv8-Pose pallet model training detection

The YOLO algorithm, a quintessential single-stage detector, treats object detection as a regression task. It divides the input image into grids, with each grid predicting bounding boxes with confidence scores based on the Intersection over Union (IoU) between predicted and ground truth boxes. For pallet pose estimation, this paper adopts an enhanced YOLOv8-Pose model to identify key points on pallets. A dataset of over 2000 pallet images, annotated with key points as shown in Figure 3, is utilized for training. The model's loss function integrates both object detection and keypoint detection losses to ensure precise pallet localization and pose estimation, with gradient descent employed for optimization.

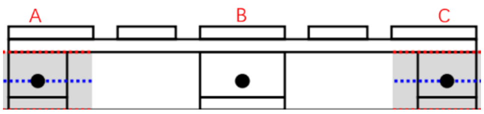


Figure 3. The marking position of key points on the pallet

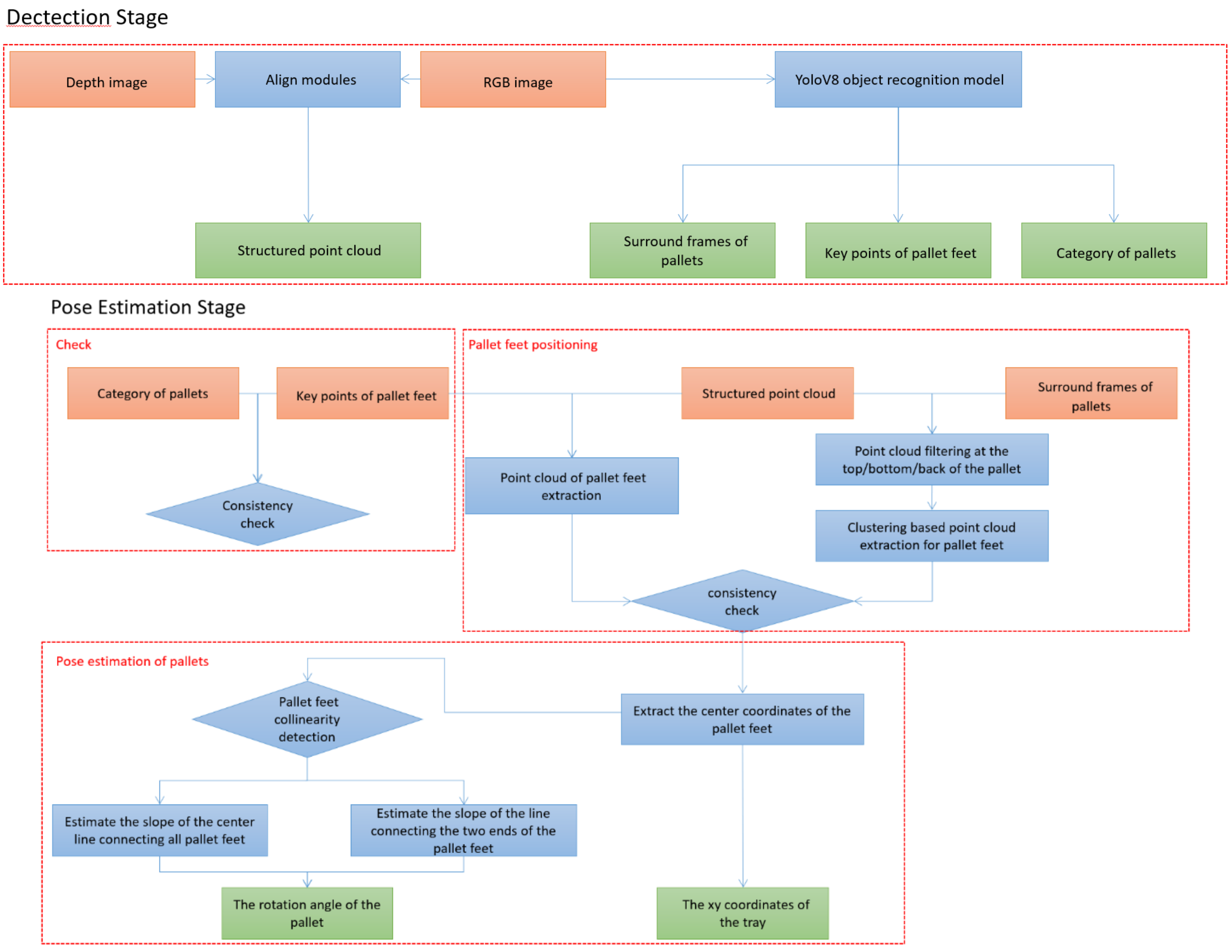


Figure 4. The holistic approach flow is divided into two parts: the pallet detection part and the pallet pose estimation part. The RGB image is detected by YOLOv8 to obtain the pallet type, detection frame and key points, and combined with the depth image to extract the point cloud at the foot of the pallet, filter out the top and bottom of the point cloud, and colinear detection at the center point, and finally obtain the position and pose of the pallet relative to the forklift.

To train the YOLOv8-Pose model for robustness and to ensure its ability to recognize various types of pallets, we have defined a loss function consisting of two main components: object detection loss and keypoint detection loss. The object detection loss assesses the difference between the predicted and ground truth boxes, ensuring the model's accuracy in locating and detecting pallets. The keypoint detection loss evaluates the model's precision in detecting key points on the pallet, which is crucial for pose estimation. The total loss L_{total} for the network is calculated as:

$$L_{total} = \lambda_{det} L_{det} + \lambda_{key} L_{key} \tag{1}$$

where L_{det} is the object detection loss and L_{key} is the keypoint detection loss, with λ being the hyperparameter. The object detection loss L_{det} is defined as:

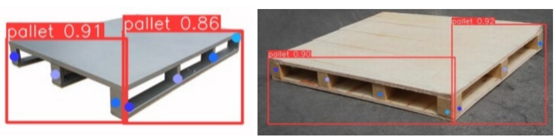
$$L_{det} = \lambda_{cls} L_{cls} + \lambda_{box} L_{box} + \lambda_{obj} L_{obj} \tag{2}$$

including classification loss L_{cls}, bounding box localization loss L_{box}, and objectness confidence loss L_{obj}.

$$L_{key} = \lambda_{kpts} L_{kpts} + \lambda_{kpts\_conf} L_{kpts\_conf} \tag{3}$$

The keypoint detection loss is further broken down into keypoint similarity loss L_{kpts} and keypoint confidence loss L_{kpts_conf}.

We utilize the trained pallet detection model, accelerated with TensorRT[13] technology, to identify and mark the position, category, and image coordinates of the pallet feet from camera-captured images, as illustrated in Figure 5. The model's output is then compared to check the consistency between the detected pallet category and the number of pallet feet. Inconsistencies may indicate partial occlusion of the pallet, which will be further examined during the subsequent pose estimation process.



(a) (b)

Figure 5. The output of the model

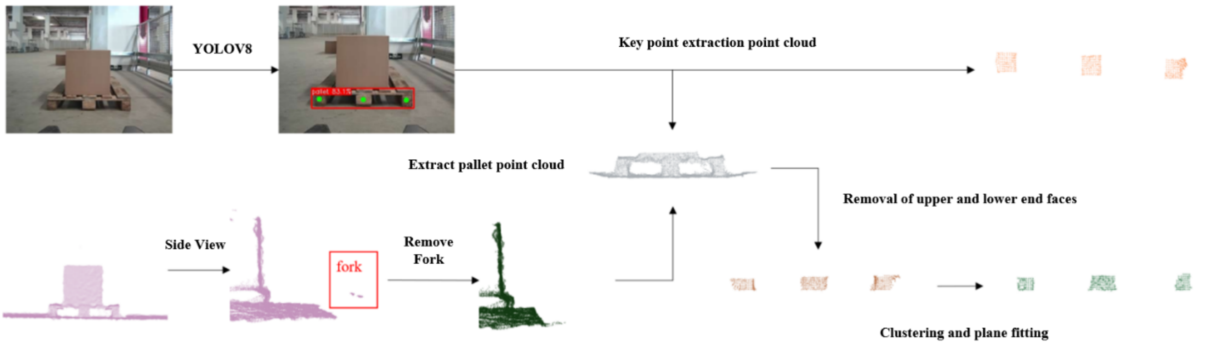


Figure 6. The extraction process of the point cloud at the foot of the pallet: one method is point cloud clustering and plane fitting, and the other method is key point extraction

B. *Pallet foot point cloud extraction*

The process for extracting the pallet foot point cloud is shown in Figure 6. Initially, the camera point cloud is obtained from the depth camera data. The point cloud of the forklift's tines is then removed from the point cloud. Subsequently, the point cloud is localized using the detection results from the YOLOv8 model, and finally, only the point cloud data of the pallet is retained.

Due to the limited precision of depth cameras, the measurement information becomes more divergent as the distance increases, such as flat surfaces being measured as curved surfaces. How to use high-variance measurement information to output accurate poses is also a challenging problem. Therefore, when using depth cameras to acquire point clouds, the quality of the point clouds is greatly affected by distance. As a result, we employ two strategies to obtain pallet foot point cloud data to ensure the effectiveness of pose estimation for pallets at a greater distance.

In the acquisition of point cloud data by depth cameras, accurately extracting the information of the pallet feet is crucial for the pose estimation of the pallet. To ensure effective acquisition of pallet foot point cloud data even at a distance, we have employed two strategies.

Firstly, based on the bounding box position information obtained from the depth camera's model inference, we filter out point cloud data related to the pallet. This step is to narrow down the processing scope and improve the efficiency of subsequent processing. Next, we remove the point cloud data of the forklift's tines to eliminate interference. Then, we perform downsampling to reduce the amount of data and filter out noise points through filtering processes, ensuring the quality of the point cloud data. By applying ground and top removal algorithms, we can further purify the point cloud data, excluding interference from parts other than the pallet. Subsequently, we conduct fast Euclidean clustering on the input point cloud data and perform plane fitting on each clustering result to determine geometric features, selecting plane point clouds with angles less than 30 degrees with the Z-axis as the true foot area of the pallet. This process ensures that the extracted point cloud data accurately reflects the real position and shape of the pallet feet.

In scenarios with multiple pallets, if the depth camera detects multiple pallets simultaneously, the algorithm will select based on the distance from the center point of the bounding box to the vehicle, prioritizing the closest pallet for processing to enhance the accuracy and efficiency of pose estimation.

In some cases, directly extracting the pallet foot from the complete point cloud may be challenging, such as when the pallet is partially occluded or the measurement precision of the depth camera is limited. To address this issue, we introduce a second strategy that relies on the keypoint position information obtained from the detection model.

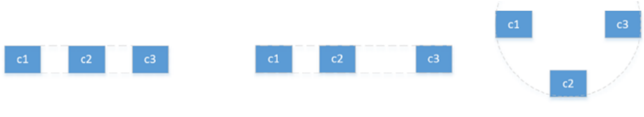
Firstly, based on the key points identified by the detection model, we extract point cloud data within a 10x10 pixel area around the pallet. This local area of point cloud data contains detailed information of the pallet feet. By downsampling these local point cloud data, we can reduce the data volume while retaining sufficient information to reconstruct the three-dimensional structure of the pallet feet.

Finally, we merge all the processed local point cloud data to form a complete object point cloud, representing the point cloud of the pallet foot area. With this method, even if the point cloud data is incomplete or has a high variance, we can still obtain a relatively accurate pallet foot point cloud, providing reliable foundational data for subsequent pose estimation.

After determining the pallet foot point cloud with these two strategies, it is then transformed from the camera coordinate system to the vehicle coordinate system.

C. *Pose estimation*

After extracting the point cloud of the pallet feet, we first use the Euclidean clustering algorithm to cluster the point cloud, resulting in three point cloud clusters corresponding to the three pallet feet. For each foot point cloud, we calculate its length, width, and height, and store the centroid information. As shown in Figure 7 for the common type of a three-foot pallet, we obtain the center coordinates of each pallet foot, which are c1, c2, and c3. We also obtain the maximum and minimum values in the horizontal direction for each foot, and the distance between adjacent feet is the pallet hole spacing. Knowing the forklift's tine width and tine spacing, we can determine whether the pallet is suitable for forklift picking based on the hole spacing.



(a) (b) (c)

Figure 7. Top view of the front end of a common tripod pallet foot

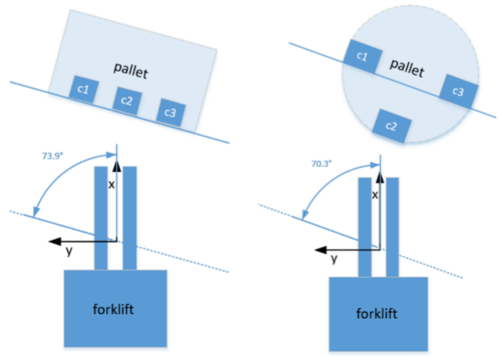
(a)Three-leg collinear center fork pallet (b)Three-leg collinear eccentric fork pallet (c)Three-leg non-collinear center fork pallet

After making the determination, we proceed with pose estimation. First, we iterate through the information of each pallet foot, extract the y-coordinates of their centroids, and sort them in ascending order, obtaining the sorted foot indices. Then, based on the sorted indices, we retrieve the center point coordinates of the left foot, middle foot, and right foot, and use the middle foot's center point as the pallet's center point.

Next, we check if the three feet are collinear. We construct a line using c1 and c3, and then measure the distance of c2 to this line. If the distance exceeds a certain threshold, we can conclude that the three feet are not collinear. If they are collinear, we use the Random Sample Consensus (RANSAC) algorithm to fit the centers of the three feet to determine the direction, calculating the angle between this line and the expected direction (x-axis), and considering the positive or negative of the direction.

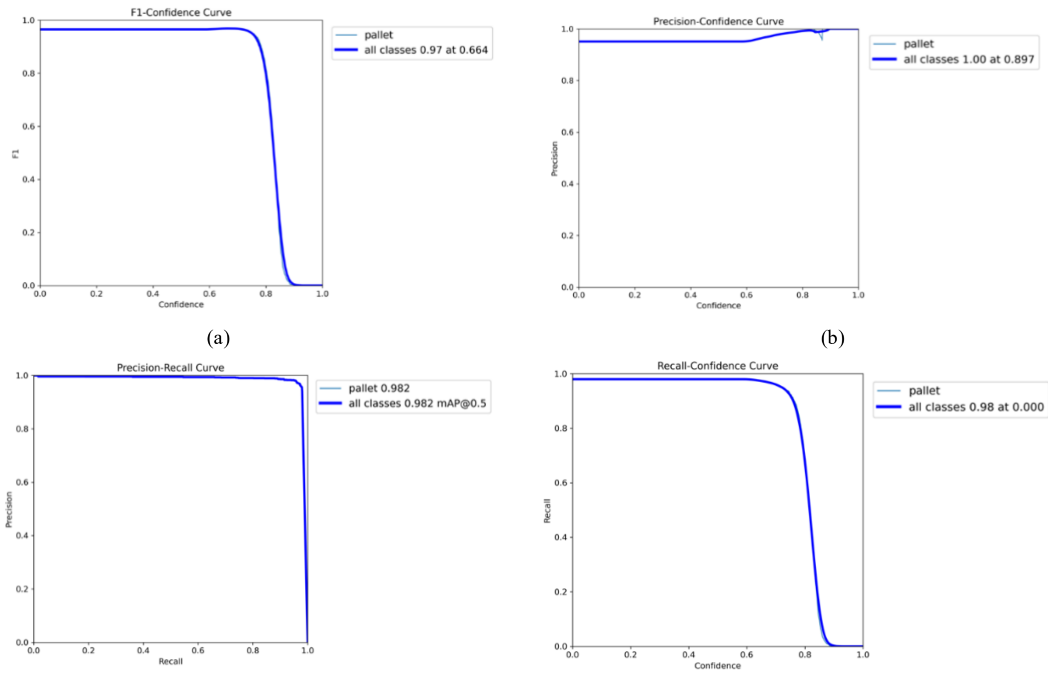
If the feet are not collinear, we calculate the angle between the vector of the right foot and left foot centroids and the expected direction vector.

In the physical world, pallets are typically placed horizontally, so there is only rotation in the horizontal direction. By defining the coordinate system with x pointing forward, y pointing to the left, and z pointing upward, the simplified schematic for the pose estimation of the pallet is as shown in Figure 8. The calculated x, y, z coordinates of the pallet's center point (ctr) and the orientation angle (theta) represent the relative position and the angle of inclination of the pallet with respect to the center of the forklift vehicle, which is the pose of the pallet relative to the forklift.



(a) (b)

Figure 8. Schematic diagram of the forklift's estimation of the pallet pose (a)Three-leg collinear (b)Three-leg non-collinear



(a) (b) (c) (d)

Figure 9. The YOLOv8-Pose model evaluates the index result

(a) F1_curve (b) P_curve (c) PR_curve (d) R_curve

IV. Experiment

A. YOLOv8 model training inference

To train the YOLOv8-Pose model, we compiled a dataset of over 2000 diverse pallet images with manual annotations, including bounding boxes and keypoint positions, to enhance the model's recognition and generalization capabilities. The dataset was split into training (80%) and validation (20%) sets for effective model training and performance evaluation.

Our trained YOLOv8-Pose model demonstrated strong performance in pallet detection tasks, accurately identifying various pallet types across different confidence thresholds. Evaluation metrics such as F1-Confidence curves, Precision-Confidence curves, Precision-Recall curves, and Recall-Confidence curves shown in Figure 9 confirmed the model's high accuracy and robustness in detecting pallets, maintaining a balance between precision and recall even at higher confidence levels.

Then, we evaluated the performance boost of TensorRT in inference speed using a dataset with multiple test samples. Fig. 10 showed a roughly 2.5x speedup compared to traditional inference, indicating improved efficiency. Future work will further optimize TensorRT for better real-time support.

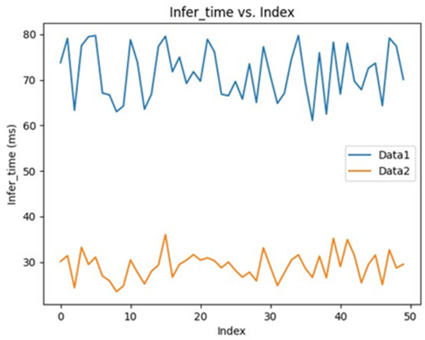


Figure 10. Comparison of inference time

B. Pose estimation accuracy

To assess the accuracy of our pose estimation algorithm, we deployed it on a forklift equipped with a depth camera through the ROS2 robotic operating system and conducted experiments in real logistics and warehousing scenarios. The algorithm's precision was evaluated by calculating the errors between the estimated and true poses of the pallet. Table 1 presents the average (D_{avg}) and maximum (D_{max}) errors in the pallet's horizontal offsets (x, y) and angular deviation (Θ) at various distances. The results indicate that our algorithm maintains errors within a small range, with an average error of 0.6cm, 0.4cm, and 0.7°, and a maximum error not exceeding 1cm, 0.5cm, and 1°, meeting our preset precision requirements and offering practical significance.

TABLE I. Error table of horizontal offset (cm) and angular offset (°) of pallet

Horizontal x	D_{avg}	D_{max}	Horizontal y	D_{avg}	Dmax	Angle Θ	Θ_{avg}	Θ_{max}
50	0.62	0.75	20	0.31	0.35	0	0.56	0.94
100	0.60	0.71	40	0.33	0.41	30	0.78	0.95
150	0.91	0.98	60	0.40	0.47	60	0.86	0.94
200	0.95	0.99	80	0.39	0.46	120	0.92	0.97
250	0.98	0.99	100	0.43	0.50	150	0.95	0.99

TABLE II. Comparison of pallet pose estimation methods

Name	Sensor	Primary Methods	Maximum error	Cost	Templates/tags
Algorithm A[3]	Laser rangefinder	Indirect measurements	±0.8cm, ±0.4°	High	True
Algorithm B[8]	Camera	ArUco tags	±3cm, ±1°	High	True
Algorithm C[14]	Camera	Geometric constraints	±10cm, ±5°	Low	False
Algorithm D[15]	ToF camera	Template matching	±1.5cm, ±3°	Low	True
Algorithm E[7]	RGB-D camera	Template matching	±3cm, ±2°	Low	True
Ours	RGB-D camera	Visual point cloud fusion	±1cm, ±1°	Low	False

Furthermore, to validate the detection performance during the forklift's movement, we maneuvered the forklift to approach the pallet from a distance and recorded the algorithm's output. According to our experiments, within a distance of 2m, the pallet's horizontal and angular deviations were stable, achieving a range of ±1cm and ±1°, fully satisfying the precision requirements for forklift pallet handling. At distances greater than 2m, due to poorer point cloud quality and increased 3D camera measurement errors, the deviations reached ±2cm and ±2° but still met the requirements for guiding the forklift towards the pallet.

In addition, as shown in Table 2, within the same effective range of the camera (2m, 45°), we compared our algorithm with several commonly used pallet pose estimation algorithms. Our algorithm has higher accuracy and lower cost respectively. It has better reliability and adaptability in practical applications.

The advantages of our proposed algorithm include not requiring manual marking on pallets, thus avoiding increased maintenance costs; supporting various pallet types with high recognition rates and robust localization; and fast model inference, enhancing real-time performance.

V. Conclusions

This paper presents a forklift pallet detection and pose estimation method based on vision and 3D point cloud. The key findings and contributions of this study are summarized as follows:

- Improved YOLOv8 Network: The improved YOLOv8 network has been utilized to achieve precise object detection and key point recognition for pallets.
- Fusion of Vision and Point Cloud Technology: By integrating 3D point cloud data to locate the pallet feet, the system's accurate estimation of the pallet's pose has been enhanced.
- High Precision and Robustness: Experimental results demonstrate that this method has high precision and strong robustness in pallet detection within industrial environments, with a recognition rate exceeding 98%.
- Minimal Error: The positioning error is effectively controlled, with distance errors kept within 1 cm and angular errors within 1°.

Although performance may decrease beyond the effective distance of the camera, the adoption of advanced depth sensing technology is expected to improve algorithm performance in the future to meet the demand for longer working distances. This research is of great value to the automation and intelligentization of forklift operation.

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